TOPS NETWORK SEGMENTATION AND ACCESS CONTROL

Cisco Packet Tracer Practical Documentation

Topic: Router Configuration, VLANs, Inter-VLAN Routing, and ACLs

Student Name: _____

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1. Aim

To design and configure a departmental network for TOPS using one router, one Layer-2 switch, three VLANs, router-on-a-stick inter-VLAN routing, and Access Control Lists (ACLs) to enforce the required communication policy.

2. Software and Devices Required

• Cisco Packet Tracer

• 1 router with a FastEthernet0/0 interface (as shown in the supplied topology)

• 1 Cisco 2960 switch

• 3 PCs: PC0 (Student), PC1 (Faculty), and PC2 (Admin)

• Copper straight-through Ethernet cables

3. Network Topology and Port Mapping

Router0 FastEthernet0/0 connects to Switch0 FastEthernet0/1. PC0 (Student) connects to Switch0 Fa0/2, PC1 (Faculty) to Fa0/3, and PC2 (Admin) to Fa0/4.

• Router0 Fa0/0 ↔ Switch0 Fa0/1 (802.1Q trunk)

• PC0 Student ↔ Switch0 Fa0/2 (access VLAN 10)

• PC1 Faculty ↔ Switch0 Fa0/3 (access VLAN 20)

• PC2 Admin ↔ Switch0 Fa0/4 (access VLAN 30)

4. IP Addressing Plan

Device / Department	IP Address	Subnet Mask	Default Gateway
PC0 — Student / VLAN 10	192.168.10.2	255.255.255.0	192.168.10.1
PC1 — Faculty / VLAN 20	192.168.20.2	255.255.255.0	192.168.20.1
PC2 — Admin / VLAN 30	192.168.30.2	255.255.255.0	192.168.30.1

5. Access Control Policy

Required source-to-destination access matrix:

Source → Destination	Students	Faculty	Admin
Students	Allow	Deny	Deny
Faculty	Allow	Allow	Deny
Admin	Deny	Allow	Allow

Note: The ACL examples demonstrate the directional policy for ICMP ping testing. Stateless ACLs do not track sessions; other applications may require protocol-specific return rules or a stateful firewall.

6. Step-by-Step Configuration

Step 1 — Build the topology

• Open Cisco Packet Tracer.

• Place one router, one Cisco 2960 switch, and three PCs.

• Rename devices if desired: Router0, Switch0, PC0-Student, PC1-Faculty, and PC2-Admin.

• Connect Router0 Fa0/0 to Switch0 Fa0/1.

• Connect PC0 to Fa0/2, PC1 to Fa0/3, and PC2 to Fa0/4 on the switch.

• Use copper straight-through cables.

Step 2 — Create VLANs and configure switch ports

Open Switch0 → CLI and enter the commands one line at a time:

```
enable
configure terminal
vlan 10
 name STUDENTS
exit
vlan 20
 name FACULTY
exit
vlan 30
 name ADMIN
exit
interface fastEthernet 0/2
 switchport mode access
 switchport access vlan 10
 no shutdown
exit
interface fastEthernet 0/3
 switchport mode access
 switchport access vlan 20
 no shutdown
exit
interface fastEthernet 0/4
 switchport mode access
 switchport access vlan 30
 no shutdown
exit
interface fastEthernet 0/1
 switchport mode trunk
 switchport trunk allowed vlan 10,20,30
 no shutdown
exit
end
write memory
```

Verify:

```
show vlan brief
show interfaces trunk
```

Step 3 — Configure router-on-a-stick

Open Router0 → CLI. These commands match the FastEthernet0/0 interface in the topology:

```
enable
configure terminal
interface fastEthernet 0/0
 no ip address
 no shutdown
exit
interface fastEthernet 0/0.10
 encapsulation dot1Q 10
 ip address 192.168.10.1 255.255.255.0
exit
interface fastEthernet 0/0.20
 encapsulation dot1Q 20
 ip address 192.168.20.1 255.255.255.0
exit
interface fastEthernet 0/0.30
 encapsulation dot1Q 30
 ip address 192.168.30.1 255.255.255.0
exit
end
write memory
```

Step 4 — Configure PC IP addresses

For each PC, open Desktop → IP Configuration and enter the IP address, subnet mask, and default gateway from Section 4. Leave DNS blank unless required.

Step 5 — Configure ACLs

Apply the ACLs inbound on each departmental router subinterface. These are basic stateless rules intended for the stated ping-direction tests; verify the results and ACL counters in Packet Tracer.

```
enable
configure terminal
ip access-list extended STUDENTS-IN
 permit icmp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 echo-reply
 deny ip 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255
 deny ip 192.168.10.0 0.0.0.255 192.168.30.0 0.0.0.255
```

```
 permit ip any any
exit
ip access-list extended FACULTY-IN
 permit icmp 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255 echo-reply
 deny ip 192.168.20.0 0.0.0.255 192.168.30.0 0.0.0.255
 permit ip any any
exit
ip access-list extended ADMIN-IN
 deny ip 192.168.30.0 0.0.0.255 192.168.10.0 0.0.0.255
 permit ip any any
exit
interface fastEthernet 0/0.10
 ip access-group STUDENTS-IN in
exit
interface fastEthernet 0/0.20
 ip access-group FACULTY-IN in
exit
interface fastEthernet 0/0.30
 ip access-group ADMIN-IN in
exit
end
write memory
```

Important: These ACLs are not a stateful firewall for every application. For broader application-level control, use stateful inspection or explicit return-traffic rules.

7. Connectivity and ACL Testing

Open each PC → Desktop → Command Prompt. Run each command from the indicated source PC:

Source PC	Command	Expected Result
PC0 Student	ping 192.168.20.2	Deny
PC0 Student	ping 192.168.30.2	Deny
PC1 Faculty	ping 192.168.10.2	Allow
PC1 Faculty	ping 192.168.30.2	Deny
PC2 Admin	ping 192.168.10.2	Deny
PC2 Admin	ping 192.168.20.2	Allow

This topology has one PC per department, so it cannot test same-department connectivity between two different PCs. Add a second PC per VLAN if required.

The first ping may fail while ARP resolves the destination MAC address. Repeat the ping and record the final result.

8. Verification Commands

On Router0

```
show ip interface brief
show ip route
show access-lists
show running-config
```

On Switch0

```
show vlan brief
show interfaces trunk
show running-config
```

Confirm router subinterfaces are up/up, VLAN 10/20/30 contain the correct access ports, Fa0/1 is trunking, and ACL counters increase when tests are run.

9. Save and Submit

• In Packet Tracer, choose File → Save As.

• Save the project as TOPS_Network_Segmentation.pkt.

• Capture a screenshot of the complete topology.

• Capture switch VLAN and trunk verification outputs.

• Capture router subinterface and ACL configuration outputs.

• Capture successful and blocked ping results.

• Submit this PDF/DOCX documentation and the Packet Tracer PKT file.

10. Result / Conclusion

The TOPS network is designed with three separate VLANs: Students (VLAN 10), Faculty (VLAN 20), and Admin (VLAN 30). Router-on-a-stick provides inter-VLAN routing, while extended ACLs enforce the specified communication restrictions. Confirm the final practical result by running the ping tests and checking ACL counters in Cisco Packet Tracer.